

HESPERO AGCM, AB, CanadaWMO: **715400**Lat: **52.310N**Lon: **114.486W**Elev: **965**StdP: **90.26**Time Zone: **-7.00 (NAM)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-31.3	-27.8	-34.5	0.2	-31.1	-31.1	0.2	-27.7	11.3	-7.1	9.9	-9.8	1.2	120	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	26.9	16.3	25.3	15.8	23.8	15.1	18.6	24.4	17.4	23.2	16.3	21.8	3.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.4	13.1	21.8	15.1	12.1	20.2	14.0	11.2	18.8	56.9	24.3	52.8	23.3	49.2	21.9	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.3	7.1	-35.8	30.7	2.5	2.0	-37.6	32.2	-39.0	33.4	-40.4	34.5	-42.3	36.0		
(5)			-35.9	21.2	2.5	1.8	-37.7	22.5	-39.1	23.5	-40.5	24.5	-42.3	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-9.4	-11.2	-5.1	2.7	9.4	13.1	15.5	14.7	10.0	3.3	-4.9	-11.2	(6)
(7)		DBStd	11.34	7.65	7.91	7.52	4.89	4.11	2.82	2.46	2.83	4.20	4.63	7.21	7.43	(7)
(8)		HDD10.0	3316	601	593	467	220	62	7	0	2	50	210	447	657	(8)
(9)		HDD18.3	5849	859	827	725	468	278	157	93	114	251	464	697	916	(9)
(10)		CDD10.0	516	0	0	0	2	42	100	169	149	49	4	1	0	(10)
(11)		CDD18.3	7	0	0	0	0	0	1	4	3	0	0	0	0	(11)
(12)		CDH23.3	432	0	0	0	2	41	39	149	137	62	2	0	0	(12)
(13)		CDH26.7	62	0	0	0	0	3	3	22	24	10	0	0	0	(13)
(14)	Wind	WSAvg	2.9	2.6	2.5	2.8	3.6	3.7	3.4	3.0	2.6	2.7	2.9	2.5	2.4	(14)
(15)	Precipitation	PrecAvg	523	23	17	22	27	63	106	91	74	41	24	22	18	(15)
(16)		PrecMax	638	58	37	48	93	124	183	211	125	86	54	70	37	(16)
(17)		PrecMin	380	3	2	2	4	32	20	26	16	4	9	7	6	(17)
(18)		PrecStd	66	14	10	12	21	22	39	42	31	25	13	15	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.3	7.8	13.1	22.6	26.9	26.2	28.8	29.4	28.2	22.1	15.2	8.2	(19)
(20)			MCWB	3.2	2.2	5.9	10.6	13.0	15.8	18.3	16.8	14.4	10.9	7.8	2.7	(20)
(21)		2%	DB	7.0	4.2	9.3	18.8	24.3	24.4	26.7	26.4	25.0	17.9	10.3	4.7	(21)
(22)			MCWB	2.5	0.5	3.7	8.8	11.8	15.0	18.2	16.8	14.2	9.1	4.6	0.6	(22)
(23)		5%	DB	4.4	2.2	6.9	14.5	21.9	22.6	24.9	24.7	22.6	15.1	6.9	1.5	(23)
(24)			MCWB	1.0	-0.5	2.4	6.1	11.1	13.9	17.1	16.4	13.5	7.9	2.8	-1.2	(24)
(25)		10%	DB	1.5	0.1	4.5	11.5	19.2	20.8	23.1	22.7	19.6	12.1	4.4	-1.0	(25)
(26)			MCWB	-0.9	-1.8	1.2	5.0	9.8	13.0	16.2	15.7	12.2	6.2	1.3	-2.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.2	6.4	10.6	13.7	17.9	20.8	19.8	16.2	12.0	8.2	3.1	(27)
(28)			MCDWB	8.2	6.8	12.3	21.8	24.2	23.8	25.7	24.7	24.6	20.6	13.8	7.4	(28)
(29)		2%	WB	2.6	1.0	4.2	8.7	12.5	16.2	19.2	18.2	14.9	10.0	5.1	0.9	(29)
(30)			MCDWB	6.7	3.8	8.7	17.7	22.2	22.2	24.8	24.2	22.5	16.7	9.2	4.3	(30)
(31)		5%	WB	1.2	-0.3	2.7	6.8	11.5	14.9	17.9	17.1	13.8	8.3	3.2	-0.9	(31)
(32)			MCDWB	4.2	1.8	6.3	13.4	20.1	20.4	23.8	23.0	21.2	14.2	6.3	1.5	(32)
(33)		10%	WB	-0.9	-1.7	1.3	5.2	10.4	13.9	16.7	16.0	12.5	6.8	1.5	-2.8	(33)
(34)			MCDWB	1.4	0.2	4.1	10.4	18.0	19.5	22.0	21.8	19.1	11.4	4.0	-1.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.7	12.3	11.9	12.0	13.9	12.5	13.0	13.8	14.4	13.3	11.4	11.0	(35)
(36)			MCDBR	14.5	13.7	14.5	19.6	19.6	16.6	17.0	18.6	21.3	19.1	14.9	14.1	(36)
(37)			MCWBR	10.3	10.3	9.5	10.5	9.0	8.4	9.6	10.1	11.1	11.0	9.8	10.6	(37)
(38)		5% WB	MCDWB	14.2	12.7	13.6	18.0	17.1	14.2	15.8	16.3	19.1	18.0	13.9	14.2	(38)
(39)			MCWBR	10.2	9.8	9.2	9.9	8.1	7.7	9.6	9.5	10.2	10.8	9.7	11.0	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.257	0.261	0.300	0.331	0.344	0.351	0.349	0.360	0.313	0.282	0.253	0.243	(40)
(41)		taud		2.355	2.431	2.399	2.361	2.377	2.382	2.419	2.367	2.476	2.555	2.491	2.351	(41)
(42)		Ebn at Noon		764	880	900	905	907	901	896	865	876	846	779	718	(42)
(43)		Edh at Noon		58	72	92	109	113	114	108	107	85	64	50	48	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.02	1.95	3.32	4.58	5.39	5.71	6.12	5.09	3.57	2.16	1.15	0.79	(44)
(45)		RadStd		0.06	0.13	0.20	0.36	0.39	0.43	0.38	0.34	0.39	0.22	0.09	0.05	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page